Testing Against A High Dimensional Alternative (Goeman et al.)

Presenter Man? - Gavin Tovar

When? - Right now

From? - Me (aka Ya Boi)

Setting The Table? - Setting

Linear Model setting stuff y'know

$$\mathbf{Y}_{(n\times 1)} = \mathbf{X}_{(n\times(p+1))}\boldsymbol{\beta}_{((p+1)\times 1)} + \boldsymbol{\epsilon}_{(n\times 1)}, \qquad \boldsymbol{\epsilon} \sim N\left(0, \sigma^2 \boldsymbol{I}\right)$$

(Note: There are p predictors of interest, plus an intercept column in the design matrix $\mathbf{X}$.)

Why Are We Here? - Motivation

The classical approach focuses on the F-test to examine

$$H_0 : \boldsymbol{\beta}_{(-1)} = 0 \quad \text{versus} \quad H_A : \boldsymbol{\beta}_{(-1)} \neq 0,$$

i.e., whether any regression coefficient (excluding intercept) is nonzero. The resulting F-statistic is

$$F = \frac{(\mathrm{RSS}_{\mathrm{Red}} - \mathrm{RSS}_{\mathrm{Full}})/p}{\mathrm{RSS}_{\mathrm{Full}}/(n-p-1)} \sim F_{p,\, n-p-1}$$

Why Are We Here? - Motivation

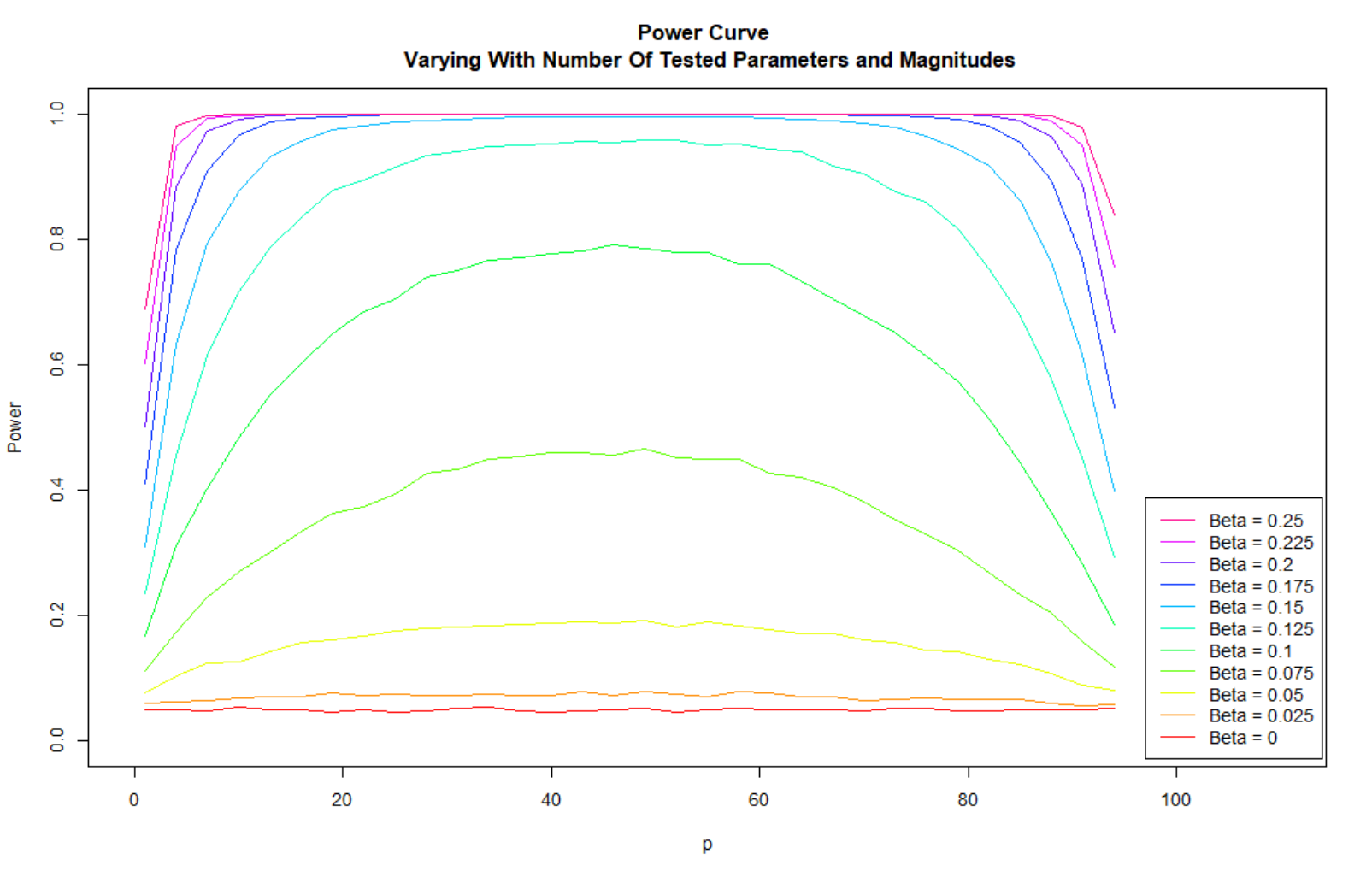

Figure 1: Basic Sim Set-up Y'know

What's Happening Here? - Method

The general formulation of their method:

(1) The idea is to concentrate the power of the test on alternatives deemed most relevant. So they're going Bayesian-style and placing a prior on $\boldsymbol{\beta}$ to accomplish this.

(2) The specific distribution on $\boldsymbol{\beta}$ is left unspecified, but the mean and covariance are parameterized as

$$E(\boldsymbol{\beta}) = \mathbf{0} \qquad E(\boldsymbol{\beta}\boldsymbol{\beta}^T) = \tau^2 \boldsymbol{\Sigma}$$

for some "well-chosen" (i.e., identity) positive (semi)-definite $p \times p$ matrix $\boldsymbol{\Sigma}$.

(3) Some details later and the authors recommend testing

$$\bar{H}_0 : \tau^2 = 0 \quad \text{versus} \quad \bar{H}_A : \tau^2 > 0$$

in the marginal model using the score test statistic

$$S = \frac{d}{d\tau^2} \log\{\bar{f}(0 \mid \mathbf{y})\}$$

Note: τ is a scale parameter reflecting the magnitude of $\boldsymbol{\beta}$

What's Happening Here? - Method

Consider the eigendecomposition of $\mathbf{X}^T\mathbf{X}$ (i.e., principal components of the design matrix). Define

$\lambda_i = i^{th}$ largest eigenvalue (the variance of the ith principal component)

$P_i =$ projection matrix onto the ith principal component direction of the design matrix.

In our linear model scenario (and with some simplifying assumptions)

$$S = \sum_{i=1}^{n} \lambda_i \frac{\mathbf{y}'\mathbf{P}_i\mathbf{y}}{\mathbf{y}'\mathbf{y}}$$

Viewed in this way, the procedure resembles a weighted principal components regression, in which evidence for association is aggregated across all principal components, weighted by the variance in each respective direction.

What's Happening Here? - Method

The authors state some Lemmas to state that the score test achieves the optimal slope of the power function against local alternatives close to the null hypothesis.

However, these optimality results depend heavily on the assumed prior structure placed on β. In particular, the choice of prior distribution effectively determines where the test will have power.

Hey Hayden's Haystack? - Analogy Time

Figure 2: Haystack for Hayden

Hey Hayden's Haystack? - Analogy Time

Figure 3: Metal Detector

Hey Hayden's Haystack? - Analogy Time

When we have $p \approx n$:

Classic F Test

Like a giant but weak metal detect

Proposed Score Test

Like a smaller but more powerful metal detector (only if the needles are oriented just right)

Scorched Earth! - Discussion

Some observations:

- Score test is not invariant (no $p > n$ test can be, but this should still give us some pause)
- Score test is biased for some alternative directions
- Score test emphasizes different directions depending on choice of $\boldsymbol{\Sigma}$
- Score test is much more variable with different realizations of design matrix $\mathbf{X}$ from a common data-generating process than is the F-test.

Scorched Earth! - Discussion

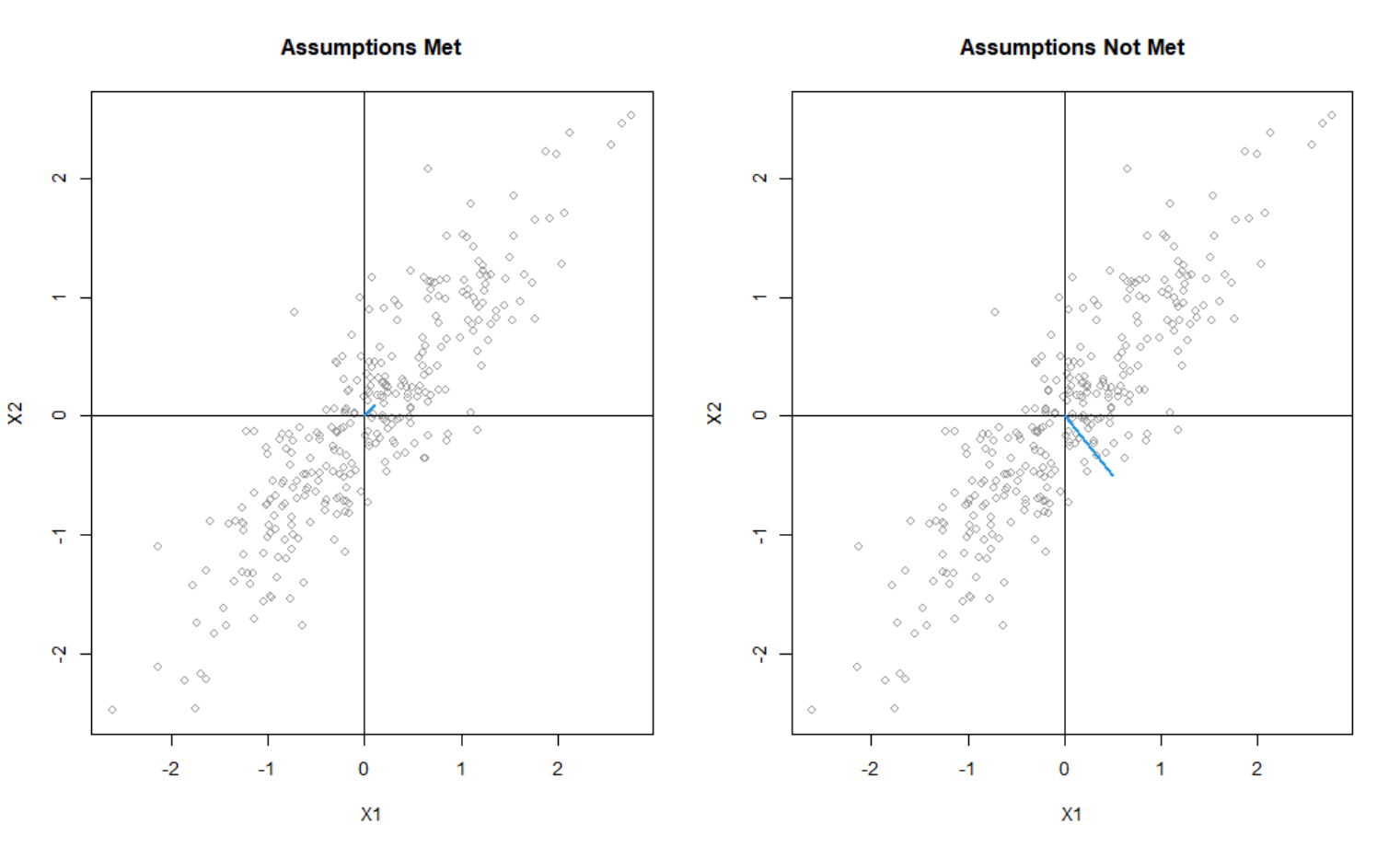

Figure 4: Illustration of alignment between predictor variation and the direction of β. The left panel shows a case where the coefficient vector aligns with the principal direction of variation in $\mathbf{X}$, while the right panel shows a case where this alignment does not occur.

Scorched Earth! - Discussion

$\boldsymbol{\beta}$	F-Test Power	Score Test Power
$\begin{pmatrix} 0.10 & 0.10 \end{pmatrix}$	0.73126	0.83634
$\begin{pmatrix} 0.50 & -0.50 \end{pmatrix}$	0.92465	0.09600

Table 1: Empirical power comparison between the classical F-test and the proposed score test under two coefficient configurations (100,000 simulations were performed to estimate the power of these tests)

Scorched Earth! - Discussion

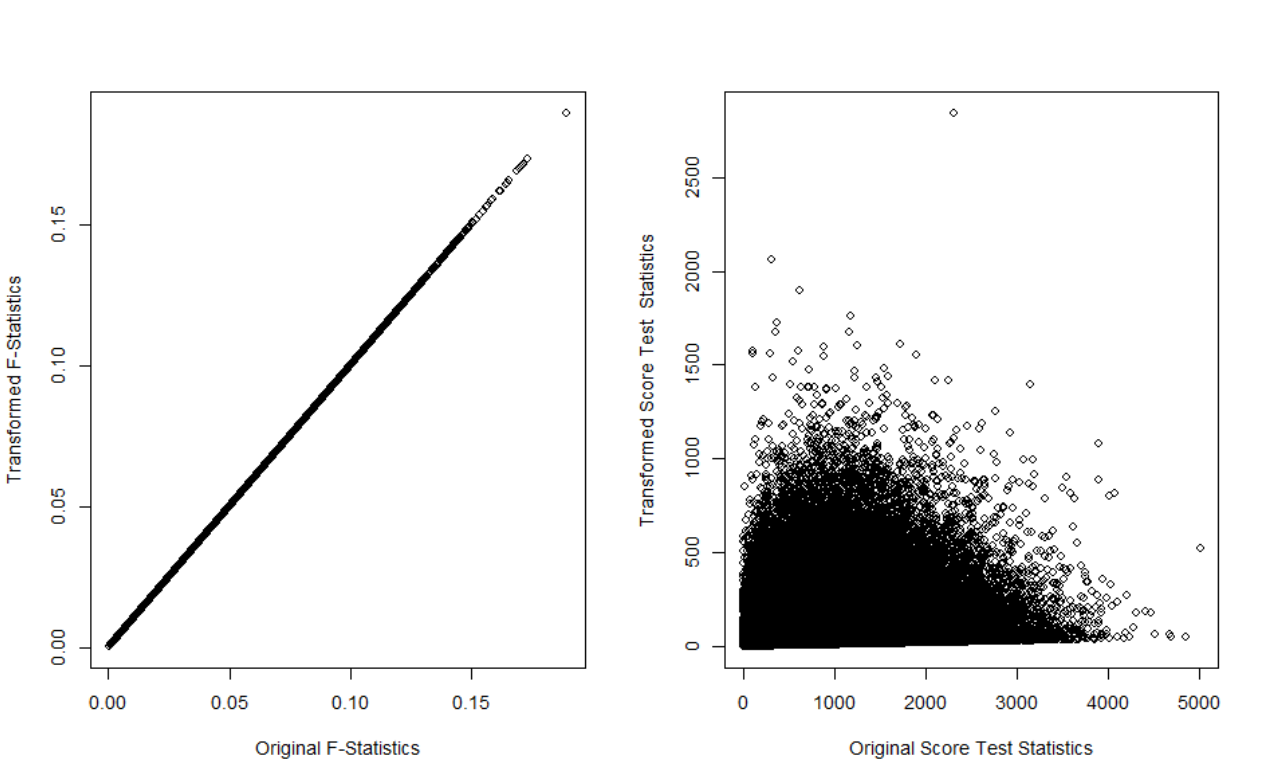

Figure 5: Comparison of the F-test and score test statistics before and after a linear transformation of the predictor space.

Scorched Earth! - Discussion

	F-Test Power	Score Test Power
$\boldsymbol{\beta} = \begin{pmatrix} 0.02 & 0.02 \end{pmatrix}$	0.81476	0.91948
$\boldsymbol{\gamma} = \begin{pmatrix} 0.20 & 0.002 \end{pmatrix}$	0.81476	0.08326

Table 2: Empirical power comparison between the classical F-test and the proposed score test under two equivalent parameterizations. Power was estimated using 100,000 simulations.

Oof! - Wrapping Up